

LÊ VĂN THUẬN

Senior AI / Software Engineer

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SUMMARY

Senior AI / Software Engineer with **7+ years** building and scaling production systems — now owning end-to-end agent-platform architecture at Vireox. Pairs deep **AI engineering** (LLM agents, multi-agent orchestration, RAG, MCP, evals, cost/latency optimization) with strong **backend foundations** (microservices, domain-driven design, high-scale data). Works at the **intersection of technology and business**, progressing toward **AI Solution Architecture**.

HIGHLIGHTS

- Own the architecture of an enterprise **AI Agent Platform** across **12 repositories** — agent backend, MCP tool gateway, RAG knowledge base, and React admin UI
- Drove a background-mode LLM report pipeline to **98%** delivery success in production, ending recurring customer-facing outages and cutting LLM operational costs **60%** through batch processing and data preprocessing
- Scaled logistics platforms to **100k+ daily transactions** at **99.9% uptime**, with **90%** improvements in API latency and report query times
- Contributed fixes **upstream to the open-source Agno agent framework**; **AWS Certified Solutions Architect – Associate**

CORE SKILLS

- **AI / LLM Engineering:** Large Language Models (LLMs) & Generative AI · LLM agents & multi-agent orchestration (Agno, LangChain, Deep-Agents) · MCP (Model Context Protocol) tool ecosystems & gateways · prompt engineering & reasoning-loop design · agent memory · LLM evals & reliability
- **RAG & Model Ops:** Retrieval-Augmented Generation (RAG) & knowledge bases (AWS Bedrock S3 Vectors, vector search, embeddings) · cost/latency optimization · LiteLLM · model providers: OpenAI, Anthropic Claude, Google Gemini (AWS Bedrock, Vertex AI)
- **Backend & Architecture:** Microservices · Domain-Driven Design · event-driven & RESTful API design · Node.js / NestJS · Python / FastAPI · .NET · GraphQL · system design · solution architecture
- **Data & Streaming:** PostgreSQL · MongoDB · Redis · vector databases · OpenSearch · Kafka · Flink · Databricks · ETL pipelines
- **Cloud & Platform:** AWS (Solutions Architect – Associate) · Kubernetes · Docker · Temporal · Celery · CI/CD (GitHub Actions, Jenkins)
- **Observability & Quality:** OpenTelemetry · Elastic APM · Sentry · LLM/agent tracing & evals · Jest · unit / e2e testing · Semgrep (SAST)
- **Frontend:** React · Next.js · Angular · TypeScript

EXPERIENCE

Senior AI Engineer | Vireox

October 2025 – Present

End-to-end ownership of the agent and product architecture for an enterprise **AI Agent Platform**, spanning **12 repositories** — a Python agent backend, data pipelines, a TypeScript MCP gateway, and a React admin UI.

- Architected the **LLM report-generation platform** (V2→V5) on OpenAI background mode with per-draft agent isolation, tracing, and Slack alerting — reaching a **98%** report delivery success rate in production and ending the recurring 15-minute request-timeout failures behind weekly customer report outages

- Built a **RAG knowledge-base platform** on **AWS Bedrock S3 Vector storage** with structured retrieval over CSV/XLSX, metadata validation, and bulk lifecycle operations — the retrieval backbone for enterprise agents
- Designed a unified **MCP (Model Context Protocol) tool gateway and catalog** — RFC 9728 per-server OAuth and dynamic discovery via a LiteLLM gateway — and contributed the underlying **MCP lifecycle-lock fix upstream to the open-source Agno framework**
- Led the **agent-runtime session-store migration from MongoDB → PostgreSQL** with forward/rollback migrations and Semgrep SAST hardening, removing a class of session-loss failures
- Hardened production reliability — fixed a **100% CPU leak** and MCP connect/close race conditions, bounded unbounded Bedrock/Vertex request timeouts, and led rapid response on urgent customer-facing report incidents
- Built supporting infrastructure end-to-end: a multi-agent **notification system** (reflection + sampling loops), an **LLM auto-evaluation framework** (migrated Celery → Temporal), OpenTelemetry observability, and the **Next.js/React admin UI**
- Pioneered **agent-driven team automation** via a Claude Code plugin marketplace and spec-driven-development harness

Software Engineer | Rennlabs

March 2024 – August 2025

Technical anchor for **CQ**, an AI-powered fundraising platform for alternative investments (deal management + investor-data enrichment).

- Led a **domain-driven modular restructuring** of the backend across 4+ repositories, cutting complexity and enabling **40% faster feature development**
- Reduced LLM operational costs **60%** via data preprocessing and batch API handling; designed a recommendation engine using hierarchical data (Itree), vector similarity, and weighted scoring
- Optimized database performance through strategic indexing and query tuning, achieving **90% reduction** in API response times
- Implemented end-to-end tracing (Sentry + OpenTelemetry/Grafana), cutting issue-detection time **90%**; built a unified web-crawler service for cross-platform data consistency
- Standardized engineering workflows (ESLint/Prettier/Husky, git flow), reached **80%** business-layer test coverage, mentored 3+ engineers, and drove AI-powered development org-wide

Technical Lead | Smartlog

December 2021 – March 2024

- Led a team of **8+ engineers** building the **Smartlog Order Management System (SOM)** — a multi-channel OMS processing **100k+ daily transactions** at **99.9% uptime**
- Architected event-driven microservices enabling seamless partner scaling; reduced order-processing time **70%** through workflow automation; integrated Shopee, Lazada, and Tiki
- Designed **Kafka/Flink ETL pipelines** for the Warehouse Management System (SWM), cutting report query time **90%** and lifting inventory accuracy to **99.5%**
- Implemented Elastic APM observability (**90%** faster debugging) and deployed AWS/Kubernetes infrastructure for high availability; mentored 5+ junior developers

Software Engineer | Smartlog

July 2020 – December 2021

- Designed backend services for warehouse and order systems processing **50k+ daily operations**; integrated 10+ e-commerce platforms via REST APIs
- Implemented Redis caching, improving application performance **60%**; mentored 2 interns and conducted code reviews

Backend Engineer | Rockship

April 2019 – July 2020

- Built backend services (Flask / PostgreSQL) and REST APIs for a flavor-navigation product; deployed and operated AWS infrastructure (EC2, S3, CloudFront)

Backend Engineer | Infinity Blockchain Labs

August 2018 – April 2019

- Developed and maintained blockchain-based applications; resolved critical bugs and shipped new features with cross-functional teams

EDUCATION & CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate**
- **B.S. Computer Science (coursework)** — Ho Chi Minh City University of Technology (2014 – 2019)

LANGUAGES

- English (working proficiency) · Vietnamese (native)